

Towards Generalization of Tactile Image Generation: Reference-Free Evaluation in a Leakage-Free Setting

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Abstract—Tactile sensing, which relies on direct physical contact, is critical for human perception and underpins applications in computer vision, robotics, and multimodal learning. Because tactile data is scarce and costly to acquire, generating synthetic tactile images provides a scalable solution to augment real-world measurements. Yet ensuring robust generalization in synthesizing tactile images—capturing subtle, material-specific contact features—remains challenging. We demonstrate overlapping train and test samples in commonly used datasets inflate performance metrics, obscuring the true generalizability of tactile models. To address this, we propose a leakage-free evaluation protocol coupled with novel, reference-free metrics—TMMD, I-TMMD, CI-TMMD, and D-TMMD—tailored for tactile generation. Moreover, we propose a vision-to-touch generation method leveraging text as an intermediate modality by incorporating concise, material-specific descriptions during training to better capture essential tactile features. Experiments on two popular visuo-tactile datasets, Touch and Go and HCT, show our approach achieves superior performance and enhanced generalization in a leakage-free setting.

I. INTRODUCTION

The tactile modality, which relies on direct physical contact to sense attributes such as force, pressure, and temperature, is a unique sensory channel that is fundamental to our understanding of the world. In computer vision and robotics, tactile cues effectively complement visual data in multimodal learning, enabling systems to discern subtle material properties that pure vision might overlook [1], [9]. This integration not only provides critical feedback for robotic manipulation through nuanced contact dynamics [19], [6], [10], but also enriches scene understanding by augmenting 3D representations with tactile insights [4]. However, obtaining robust and generalizable models remains a significant challenge across various tactile sensing tasks due to the expensive physical data collection process and non-standardized sensor outputs [16].

Tactile generation stands out as a promising avenue for synthesizing tactile images from visual and multimodal inputs, a potential solution to mitigate the scarcity of tactile data [18], [4], [15], [11]; however, evaluation remains problematic due to data leakage and lack of specialized metrics.

Data leakage poses a critical obstacle to advancing tactile sensing research, undermining the goal of robust generalization by compromising many prevalent tactile datasets [17], [5], [9]. These datasets, often derived from continuous video recordings of object interactions, inadvertently include overlapping or near-duplicate samples across training and test splits. This leakage enables models to simply memorize redundant tactile patterns. It also results in artificially inflated performance metrics across various tactile-related tasks that

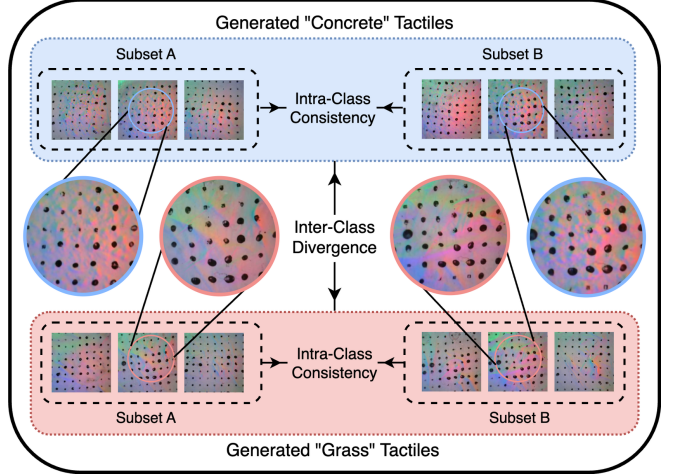


Fig. 1. An illustration of how our reference-free metrics evaluate generated tactile images across different classes. Within each class (e.g., “Concrete” or “Grass”), samples are split into two subsets (A and B) to measure *intra-class consistency*, while *inter-class divergence* quantifies how distinct the generated samples are between classes.

do not reflect real-world robustness. As a first contribution, we analyze the leakage phenomenon and introduce a leakage-free evaluation protocol that rigorously separates training and test subsets, enabling more reliable and generalizable results, meaning consistent performance in diverse, unseen scenarios.

In evaluating tactile image generation, many works have adopted conventional metrics—such as SSIM, PSNR, LPIPS, and FID—that were originally designed for natural images. These metrics often fail to accurately capture tactile-specific nuances on generated tactile images. They exhibit excessive sensitivity to irrelevant background artifacts due to passive sensor state, in the absence of contact to a material. Meanwhile they neglect fine, material-specific features that are essential for tactile perception, and obscure the true generalizability of the models. As a second contribution, we introduce a set of novel, reference-free evaluation metrics—namely, TMMD, I-TMMD, CI-TMMD, and D-TMMD—which are specifically designed to capture material-specific tactile features and eliminate the need for reference images. By leveraging a dedicated tactile encoder, these metrics quantify both intra-class consistency (ensuring internal stability) and inter-class divergence (ensuring diversity across materials) as illustrated in Fig. 1, offering accurate assessment of tactile image fidelity and diversity of generated images.

Finally, we propose a method that leverages text as an intermediate modality in the vision-to-touch generation task.

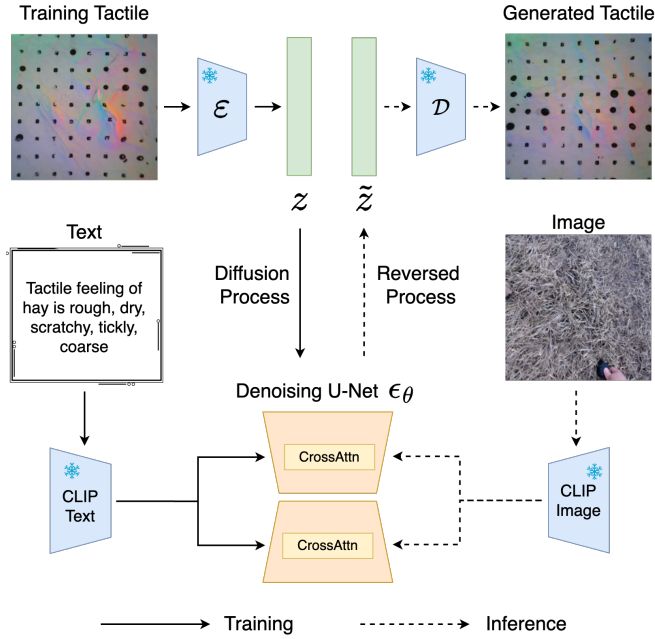


Fig. 2. Overview of our latent diffusion-based tactile image generation pipeline: visual inputs are enriched with text cues during training to guide the generation of tactile images via latent diffusion. At inference, only the visual image is required.

This is achieved by incorporating concise, material-specific textual descriptions during training. By leveraging text to emphasize essential tactile features and filter out irrelevant visual details, our method accurately captures intrinsic contact patterns and material properties, thereby enhancing the quality of the generated tactile images. We use our robust evaluation framework, which operates under an enforced leakage-free setting and utilizes novel metrics tailored specifically for tactile generation, to verify the effectiveness of our method, ensuring that the results are reliable and generalizable.

II. METHOD

Due to the labor-intensive process of collecting tactile data, we propose an approach that generates tactile images from visual inputs to alleviate data scarcity. We leverage textual descriptions of tactile properties as an intermediate bridge between vision and tactile during training. To enhance evaluation in tactile image generation, we propose several reference-free metrics designed to assess the quality, consistency, and diversity of generated tactile images without requiring ground truth comparisons.

A. Tactile Image Generation with Latent Diffusion

We employ a diffusion-based approach [12] that progressively refines random noise into coherent samples by reversing a predefined Markov chain. A triplet of visual, tactile, and background tactile images $\{x_{vis}, x_{tac}, x_{bg}\} \in \mathbb{R}^{H \times W \times 3}$ is provided. The background tactile image x_{bg} records the sensor baseline state in the absence of material contact. To guide training, we incorporate concise textual descriptions x_{tex} that emphasize essential material-specific patterns—e.g., “rough,

bumpy surface typical of concrete.” These cues help the model focus on tactile-relevant features while filtering out irrelevant visual details such as color, lighting or unrelated objects in the scene. Leveraging CLIP’s encoders, both visual and textual inputs are projected into a unified feature space, ensuring consistent representations. Our method synthesizes tactile images using only visual data at inference (Fig. 2).

The tactile image x_{tac} is first encoded into a compressed latent $z_{tac} = \mathcal{E}(x_{tac}) \in \mathbb{R}^{h \times w \times 3}$ via a pretrained, frozen VQ-GAN encoder $\mathcal{E}$ [3]. By choosing $h \times w \ll H \times W$, the aim is to preserve essential information while reducing computational costs. Next, a diffusion step $t \in \{1, \dots, T\}$ is sampled uniformly, and Gaussian noise is added according to a scheduled variance, resulting in the noisy latent z_{tac}^t . A time-conditional U-Net ϵ_θ [13] is trained to predict and remove this noise while conditioning on a frozen CLIP text encoder $\mathcal{C}_{tex}$. The training objective is:

$$\mathcal{L}(\theta) = \mathbb{E}_{z_{tac}, x_{tex}, \epsilon, t} \left[\|\epsilon_t - \epsilon_\theta(z_{tac}^t, t, \mathcal{C}_{tex}(x_{tex}))\|_2^2 \right], \quad (1)$$

where ϵ_t denotes the noise injected at time t .

During inference, the model samples random noise $z_{tac}^T \sim \mathcal{N}(0, 1)$ in latent space. The trained model ϵ_θ then denoises this latent iteratively, stepping backward from $t = T$ down to $t = 0$. Note that z_{tac}^0 represents the fully denoised final latent. Since textual cues are used only during training, the model conditions visual images with CLIP image encoder $\mathcal{C}_{vis}(x_{vis})$, combined with an unconditional pass via classifier-free guidance for better quality and diversity. The final latent z_{tac}^0 is then reconstructed by VQ-GAN decoder $\mathcal{D}$, producing generated tactile image $\hat{x}_{tac} = \mathcal{D}(z_{tac}^0)$.

B. Reference-Free Tactile Generation Metrics

Fréchet Inception Distance (FID) [7] evaluates generative models by comparing feature distributions, but it assumes a single Gaussian for high-dimensional features—overlooking complexities like multimodality or skewness—and requires around 50,000 images [7] for stable estimates, poses a significant challenge for domains like tactile sensing where data is inherently limited. In contrast, CLIP Maximum Mean Discrepancy (CMMD) [8] addresses these issues by using MMD which directly compares features without restrictive assumptions, offering a sample-efficient, unbiased, and more reliable evaluations. However, CMMD relies on CLIP embeddings, which, despite their strong semantic alignment for natural images, are not trained on tactile data.

To establish a robust evaluation metric for comparing tactile distributions, a dedicated touch encoder is required to effectively capture the underlying structure of tactile data. We achieve this by pre-training a cross-modal visual-tactile encoder using self-supervised contrastive learning. Building on prior work in contrastive pre-training [14], [5], we employ tactile-textual, vision-textual and tactile-vision contrastive losses to align tactile features with other modalities. To facilitate this alignment, the CLIP image encoder is randomly initialized and repurposed as the tactile encoder $\mathcal{C}_{tac}$, ensuring a modality-aware representation of tactile data.

1) **Tactile Maximum Mean Discrepancy (TMMD)**: Given two sets of feature embeddings $G = \{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_m\}$ and $R = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n\}$ where G denotes the generated tactile image features and R corresponds to the reference tactile image features. TMMD replaces CLIP embeddings with a dedicated tactile encoder $\mathcal{C}_{\text{tac}}$, ensuring more accurate capture of material-specific tactile nuances.

$$\widehat{\text{dist}}_{\text{TMMD}}^2(G, R) = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m k(\mathbf{g}_i, \mathbf{g}_j) + \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n k(\mathbf{r}_i, \mathbf{r}_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(\mathbf{g}_i, \mathbf{r}_j). \quad (2)$$

where k denotes a positive definite kernel measuring similarity between feature embeddings. Following [8], the Gaussian RBF kernel $k(\mathbf{x}, \mathbf{y}) = \exp(-\|\mathbf{x} - \mathbf{y}\|^2 / 2\sigma^2)$ is used to ensure TMMD effectively captures the distributional discrepancy in a high-dimensional space. This metric provides a robust comparison of tactile representations while effectively capturing complex textural details and subtle deformation patterns in the feature space. A lower TMMD score indicates better alignment between the generated and reference distributions, signifying higher quality and fidelity in the generated tactile images. Building on this foundation, we introduce reference-free metrics in the following sections to further evaluate the generated tactile images without relying on reference data.

2) **Intra TMMD (I-TMMD)**: In generative modeling, obtaining large and representative reference datasets for computing evaluation metrics is often challenging, particularly in specialized domains such as art generation, medical imaging, or tactile generation. These fields frequently suffer from data scarcity and high acquisition costs to obtain high-quality reference distributions. In such scenarios, reference-free evaluation methods become indispensable. To address this need, we introduce I-TMMD as a metric to quantify the internal consistency of generated distributions. Specifically, I-TMMD operates by splitting the set of generated samples $\mathbf{G}$ into two disjoint subsets $\mathbf{G}_1$ and $\mathbf{G}_2$ and computing the TMMD distance between them:

$$\text{I-TMMD}(\mathbf{G}) = \widehat{\text{dist}}_{\text{TMMD}}^2(\mathbf{G}_1, \mathbf{G}_2). \quad (3)$$

By evaluating the similarity between subsets, I-TMMD provides insight into the model's internal coherence and stability. I-TMMD posits that a well-trained generator, converging to a stable and well-defined distribution, should produce statistically similar, internally consistent outputs across its subsets. A lower I-TMMD score indicates greater internal consistency, and a stable, well-converged generator.

3) **Class-Aware Intra TMMD (CI-TMMD)**: While I-TMMD measures overall internal consistency by splitting the generated samples into two subsets, it does not account for class-specific variations. This can be problematic in scenarios where the generator produces samples that are consistent globally but fail to capture the distinct characteristics of

individual classes. To address this, CI-TMMD extends I-TMMD by enforcing consistency within each class. Let $\mathbf{G} = \{\mathbf{G}^1, \dots, \mathbf{G}^C\}$ denote generated tactile features partitioned into C classes. For each class c , $\mathbf{G}^c$ is split into disjoint equal subsets $\mathbf{G}_1^c$ and $\mathbf{G}_2^c$. CI-TMMD computes the average TMMD distance across classes:

$$\text{CI-TMMD}(\mathbf{G}) = \frac{1}{C} \sum_{c=1}^C \widehat{\text{dist}}_{\text{TMMD}}^2(\mathbf{G}_1^c, \mathbf{G}_2^c), \quad (4)$$

ensuring the generator preserves class-specific characteristics rather than merely achieving global uniformity.

In cases of mode collapse—where the generator produces similar outputs across classes—the resulting subsets ($\mathbf{G}_1$ and $\mathbf{G}_2$ or $\mathbf{G}_1^c$ and $\mathbf{G}_2^c$) will yield artificially low scores despite catastrophic collapse. Thus, while I-TMMD and CI-TMMD effectively assess internal consistency, they should be paired with a metric that evaluates diversity.

4) **Diversity-Aware TMMD (D-TMMD)**: To ensure classes remain separable, we propose D-TMMD, a reference-free metric that captures both intra-class similarity and inter-class divergence, complementing the internal consistency evaluations of I-TMMD and CI-TMMD. For a generator producing outputs partitioned into C classes $\mathbf{G} = \{\mathbf{G}^1, \dots, \mathbf{G}^C\}$, we construct a divergence matrix $\mathbf{D} \in \mathbb{R}^{C \times C}$ as follows:

1. **Intra-Class Consistency**: For diagonal entries ($c = c'$), split $\mathbf{G}^c$ into two subsets $\mathbf{G}_1^c$ and $\mathbf{G}_2^c$, then compute:

$$\mathbf{D}_{c,c} = \widehat{\text{dist}}_{\text{TMMD}}^2(\mathbf{G}_1^c, \mathbf{G}_2^c). \quad (5)$$

2. **Inter-Class Divergence**: For off-diagonal entries ($c \neq c'$), compute the divergence between full class sets:

$$\mathbf{D}_{c,c'} = \widehat{\text{dist}}_{\text{TMMD}}^2(\mathbf{G}^c, \mathbf{G}^{c'}). \quad (6)$$

The D-TMMD metric aggregates these values into a normalized measure of diversity:

$$\text{D-TMMD}(\mathbf{G}) = \frac{1}{C} \sum_{c=1}^C \frac{\mathbf{D}_{c,c}}{\sum_{c'=1}^C \mathbf{D}_{c,c'}}. \quad (7)$$

Here, $\mathbf{D}_{c,c}$ measures intra-class consistency, while $\sum_{c'=1}^C \mathbf{D}_{c,c'}$ captures the total divergence of class c from all classes. A low D-TMMD score indicates strong diversity, as intra-class consistency is small relative to inter-class divergence. Conversely, a high score suggests poor diversity, signaling potential mode collapse or class confusion.

III. RESULTS

A. Data Leakage in Visual-Tactile Datasets

We use two diverse visuo-tactile datasets. **Touch and Go** [17] uses a GelSight sensor; it comprises 13,900 image-touch pairs, representing 3,971 unique objects across 20 material categories.¹ **Human Collected Tactile (HCT)** [5] uses a DIGIT sensor and comprises 39,154 pairs of in-contact frames with tactile texture descriptions. Our analysis reveals

¹We leverage Molmo [2], a large vision-language model, to generate text descriptions for each sample that capture tactile-relevant visual properties.

Dataset	Material Acc (%)	Tactile Generation			
		TMMD ($\downarrow$)	I-TMMD ($\downarrow$)	CI-TMMD ($\downarrow$)	D-TMMD ($\downarrow$)
TAG-ORIG	54.7	0.33	0.04	0.21	0.06
TAG-NOLEAK	37.8	1.37	0.15	1.43	0.44

TABLE I. Comparison of material classification and tactile generation performance across TAG-ORIG and TAG-NOLEAK. The results highlight a significant inflation of metrics due to data leakage in TAG-ORIG.

Metrics	Comparison 1		Comparison 2	
	OURS	GVST [18]	OURS	OURS w/ BG
SSIM ($\uparrow$)	0.49	0.49	0.49	0.54
PSNR ($\uparrow$)	15.5	15.4	15.5	16.0
LPIPS ($\downarrow$)	0.53	0.51	0.53	0.43
FID ($\downarrow$)	43.9	43.6	43.9	46.9
CMMD ($\downarrow$)	2.52	2.49	2.52	2.44
TMMD ($\downarrow$)	1.37	1.98	1.37	2.39
I-TMMD ($\downarrow$)	0.15	0.77	0.15	1.24
CI-TMMD ($\downarrow$)	1.43	2.97	1.43	5.33
D-TMMD ($\downarrow$)	0.44	0.92	0.44	1.40
Human Pref ($\uparrow$)	75.0%	25.0%	82.5%	17.5%

TABLE III. Comparison of methods (OURS vs. GVST [18] and OURS vs. OURS w/ BG) using both metrics and human preference in TAG-NOLEAK dataset. While the newly introduced TMMD, I-TMMD, CI-TMMD and D-TMMD metrics consistently favor the same method that humans prefer, traditional pairwise (SSIM, PSNR, LPIPS) and distribution-based (FID, CMMD) metrics often diverge from human preference. Within each comparison block, the best performer for each metric is in **bold**.

systematic data leakage due to temporal proximity between training and test samples, e.g. frame 170 in testing after frame 169 in training (HCT), and test frames only 85 frames apart from training frames (Touch and Go), creating near-duplicate frames. We refer to the original splits provided by [17] and [5] as **TaG-Orig** and **HCT-Orig**, respectively. We create new splits, named **TaG-NoLeak** and **HCT-NoLeak**, where all data from a single video is exclusively allocated to either the training or test set, ensuring no overlap. Note that other datasets (SSVTP [9] and TaRF [4]) exhibit similar problems.

The analysis in Tables I and II underscores the impact of data leakage on performance metrics for different tasks in visual-tactile datasets. Table I shows material classification accuracy declining from 54.7 in TAG-ORIG to 37.8 in TAG-NOLEAK (30.9% drop), and tactile generation errors escalating, with TMMD scores increasing (thus performance decreasing) from 0.33 to 1.37, I-TMMD from 0.04 to 0.15, CI-TMMD from 0.21 to 1.43 and D-TMMD from 0.06 to 0.44. Similarly, the original split HCT-ORIG exhibits highly inflated retrieval accuracies due to leakage, with tactile-text top-1 retrieval accuracy plummeting from 36.7 to 4.1 (88.8% drop) and tactile-vision top-1 retrieval accuracy dropping from 79.5 to 7.9 (90.1% drop) in the corrected HCT-NOLEAK split (Table II). Tactile generation errors rise, with TMMD from 0.46 to 2.12 and with I-TMMD metrics from 0.13 to 0.39. These dramatic performance drops reveal that leakage inflates metrics, which we address by proposing leakage-free datasets—for trustworthy tactile model evaluation.

Dataset	Tactile-Text Retrieval		Tactile-Vision Retrieval		Tactile Generation	
	Top-1 (%)	Top-5 (%)	Top-1 (%)	Top-5 (%)	TMMD ($\downarrow$)	I-TMMD ($\downarrow$)
HCT-ORIG	36.7	70.3	79.5	95.7	0.46	0.13
HCT-NOLEAK	4.1	17.4	7.9	27.4	2.12	0.39

TABLE II. Comparison of tactile-text and tactile-vision retrieval accuracy along with tactile generation performance on the HCT dataset. The results indicate a noticeable performance drop on the HCT-NOLEAK.

B. Human Preference: Evaluation Protocol

We randomly selected 100 samples from the test set for each comparison. 5 human raters were asked, in each trial, to choose the generated tactile image that most closely resembled the reference tactile image. They were asked to focus on material-specific patterns—line-shaped/elongated textures for grass, pebbly patterns for concrete, and woven/repetitive structures for fabric—while not focusing on sensor’s default background details that appear even when a sensor does not have any contact with a material. Raters were given explicit instructions along with 5 example trials to clarify evaluation expectations. These 5 were later reused as attention checks for the annotators, with a 96% success rate, confirming rating reliability. Across raters, standard deviation was 5.3%/7.0% for Comparison 1/2, suggesting strong inter-rater consistency.

C. Novel Metrics and Comparison with Prior Art

We validate our newly introduced metrics, in the TAG-NOLEAK setting. We conducted two comparisons: (1) OURS vs GVST [18] (which uses only visual inputs for training and inference), and (2) OURS vs OURS w/ BG (which inputs background images x_{bg} (referred to as “gel status” in [15]) following to prior work [15], [4]), in Table III. We show both traditional and our novel metrics.

Our findings in Table III reveal a strong preference for OURS over both GVST and OURS w/ BG: OURS **outperforms GVST by a substantial margin** (78.8% vs. 21.2%) and surpasses OURS w/ BG even more decisively (85% vs. 15%). Notably, **this preference is clearly captured by our proposed TMMD-based metrics**, which consistently give lower (better) scores to OURS (e.g., TMMD drops from 1.98 with GVST to 1.37 with OURS, and from 2.39 with OURS w/ BG to 1.37 with OURS), aligning closely with human ratings. In contrast, FID and CMMD exhibit only marginal differences—sometimes even favoring GVST or OURS w/ BG—thereby failing to reflect the strong human preference for OURS. This discrepancy arises because FID and CMMD rely on RGB-trained encoders, limiting their ability to capture tactile-specific features.

On TAG-NOLEAK (where class labels are available), OURS lowers TMMD from 1.98 to 1.37, I-TMMD from 0.77 to 0.15, CI-TMMD from 2.97 to 1.43, and D-TMMD from 0.92 to 0.44—indicating marked improvements in fidelity, internal consistency, and class separation. Similarly, on HCT-NOLEAK (not shown in table), OURS reduces TMMD from 2.67 to 2.12 and I-TMMD from 0.61 to 0.39, further underscoring the benefit of incorporating textual cues. Overall, our method consistently achieves lower scores, demonstrating its superior ability to preserve material-specific details and generate diverse tactile outputs compared to GVST.

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